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# Paleomagnetism of the world-class Century Zn–Pb–Ag deposits, Australia

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ABSTRACT

Paleomagnetic results are reported from the Century Zn–Pb–Ag SEDEX deposits in northwestern Queensland, Australia. The stratiform mineralization occurs in fine parallel lamellae in ~1595 Ma siltstones and black shales of the Lawn Hill Formation in the Proterozoic McNamara Group. Galena has given a Pb/Pb model age of ~1570 Ma. Paleomagnetic analysis of 333 specimens from ore zones (15 sites), hanging wall sandstones or siltstone (4 sites), and footwall siltstones (5 sites) using mostly thermal and then alternating field step demagnetization, isolates a stable characteristic remanent magnetization (ChRM) for ore sites only. Step demagnetization and rock magnetic analyses of ore, Zn and Pb concentrates and tailings show that the main remanence carriers are single- or pseudosingle-domain inclusions of titanomagnetite in sphalerite and gangue, and pyrrhotite in galena with modern goethite and/or hematite from the weathering of siderite. A paleomagnetic fold test using the ChRM directions of the ore sites is positive, showing that the ore ChRM predates the $D_2$ deformation of the ~1595 to 1500 Ma Isan orogeny. The orogeny folded the main-stage mineralization, indicating that the ore retains a primary magnetization. The optimum 80% tilt-corrected unit mean ChRM direction for the ore yields a Mesoproterozoic paleopole at 1558 ± 4 Ma on the northern Australian apparent polar wander path. Thus this result both constrains the timing of mineralization and provides an upper age limit for $D_2$ deformation in the orogeny.

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## 1. Introduction

The Century Zn–Pb–Ag deposit is located on the Lawn Hill platform in northwestern Queensland, Australia, at 18.7°S, 138.6°E (Fig. 1). It is a world-class Zn–Pb–Ag sedimentary exhalative (SEDEX) deposit, containing 118 million tonnes (Mt) of ore with grades of 10.2% Zn, 1.5% Pb and 36 g/t Ag (Waltho and Andrews, 1993; Broadbent et al., 1998). The Century deposit is one of five giant SEDEX deposits (Century, HYC, Mt. Isa, George Fisher and Hilton) in the Mt. Isa–McArthur basin system (Fig. 1), which exhibit many similar geological and geochemical features, such as locations close to synsedimentary faults, laminated bedding-parallel sulphide minerals in black shales, and broad ranges of $\delta^{34}S$ values for the sulphide minerals (Large et al., 2005). Various genetic models – syngenetic through diagenetic to epigenetic models – have been proposed for their origin. However, the lack of suitable minerals for high-precision radiometric age dating has led to a lack of agreement on a given model. Symons (2007) reported paleomagnetic results for the HYC Zn–Pb deposit in the McArthur Basin and concluded controversially that the HYC deposit had an epigenetic origin. This study extends the paleomagnetic approach to studying such SEDEX deposits and provides an estimate for the mineralization age of the Century Zn–Pb–Ag deposit.

## 2. Geology

The stratiform mineralization in the Century deposit occurs as fine parallel lamellae in a 40 m-thick interlayered black shale and siltstone sequence belonging to the ~1595 Ma Pmh4 unit of the Proterozoic Lawn Hill Formation in the upper part of the McNamara Group of the Lawn Hill platform in the Western Fold Belt of the Mount Isa district (Figs. 1, 2) (Andrews, 1998; Page et al., 2000). The McNamara Group was deformed by the 1595–1500 Ma Isan orogeny, which is comprised of three main deformation stages, $D_1$, $D_2$ and $D_3$ (Blake et al., 1990; O'Dea et al., 1997; Broadbent et al., 1998). Local $D_1$ southeast to northwest compression appears to be synchronous with the main-stage mineralization (Ord et al., 2002). The absolute age of the $D_1$ phase of the Isan orogeny is poorly constrained, but metamorphic zircon overgrowths in the Eastern Fold Belt of the Mount Isa district have been interpreted to have formed at the peak of the $D_1$ phase at 1584 ± 17 Ma (Page, 1993). $D_1$ in the Century area is indirectly correlated with the final stages of the $D_1$ event of the Isan Orogeny (Ord et al., 2002). $D_2$ is an east–west compression event that includes folding of the main-stage mineralization and is interpreted to have formed at a late stage of the Isan orogeny (Broadbent et al., 1998).

Century is located adjacent to a regional synsedimentary fault, the Termite Range fault (Figs. 3, 4), that is thought to have channeled

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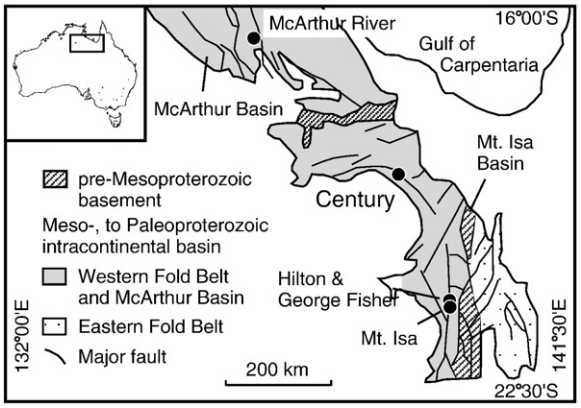



**Fig. 1.** Location map for the Century mine and other major Zn–Pb–Ag SEDEX mines in the Mt. Isa–McArthur basin of northern Australia (modified from Large et al., 2005).

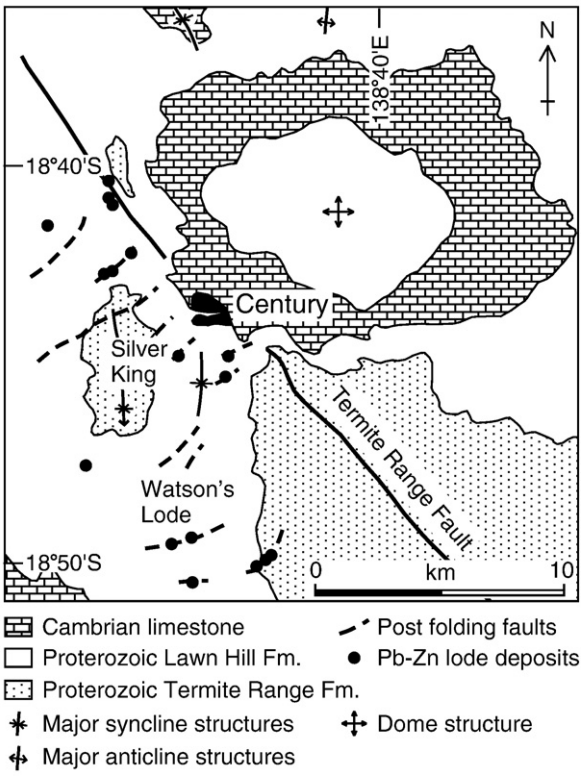



**Fig. 3.** Simplified geology of Lawn Hill region, showing the location of the Century Zn–Pb–Ag deposit and smaller crosscutting faults with Pb–Zn lodes in relationship to the Termite Range fault (modified from Large et al., 2005).

fluids into the ore environment (Broadbent et al., 1998). The Century ore body comprises two major blocks, the northern and southern blocks, which are separated by the north-dipping normal Pandora's fault (Waltho and Andrews, 1993; Broadbent et al., 1998) (Fig. 4).

The Mineralization Sequence (Fig. 2) is divided into four main units: a) a series of interbedded stylolitic and sideritic siltstones and mineralized black shales that is up to 25 m thick with individual shale and siltstone layers varying in thickness between 0.1 and 1.5 m (Unit 100); b) a black shale package about 4 to 5 m thick that is characterized by fine sphalerite and pyrite lamellae (Unit 200); c) an ~5 m siltstone-dominated unit that is comprised of interbedded black shale-siltstone layers and stylolitic and sideritic siltstone layers (Unit 300); and, d) a shale-dominated unit that is comprised of a 15 m-thick mineralized, finely laminated black shale sequence (Unit 400) (Table 1) (Waltho and Andrews, 1993). Sulphide mineralization is concentrated dominantly in the shale beds, and weakly observed in the siltstone facies. Sphalerite is the most common sulphide with minor pyrite and galena. Chalcopyrite is rare in the main-stage mineralization (Broadbent et al., 1998). The shale and siltstone are composed of fine-grained detrital quartz, illite and carbonaceous matter. The clastic components of the shale and siltstone are very similar. The main differences are their grain size, scale of layering and the amount of the incorporated organic matter, *i.e.* the shale facies has as much as 5% total organic carbon within the mineralization. The sphalerite is associated with significant quantities of pyrobitumen. Sideritic carbonate with ~70% Fe is the principal iron-bearing gangue phase (Broadbent et al., 1998).

Polito et al. (2006) reported fluid inclusion homogenization temperatures between 74 °C and 125 °C from coarse-grained sphalerite in the ore zone, which are lower temperatures than the 120 °C to

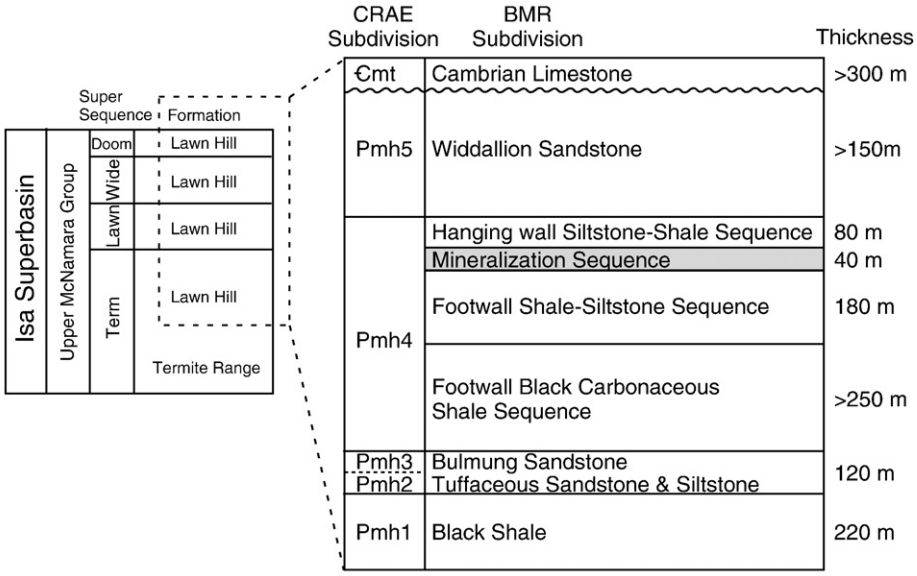



**Fig. 2.** A schematic diagram of the Lawn Hill Formation stratigraphy, Century area (modified from Waltho and Andrews, 1993; Feltrin, 2008). At Century, unit Pmh6 is entirely missing below the Cambrian unconformity. BMR = Bureau of Mineral Resources, Geology and Geophysics; CRAE = CRA Exploration Pty Limited (Waltho and Andrews, 1993).

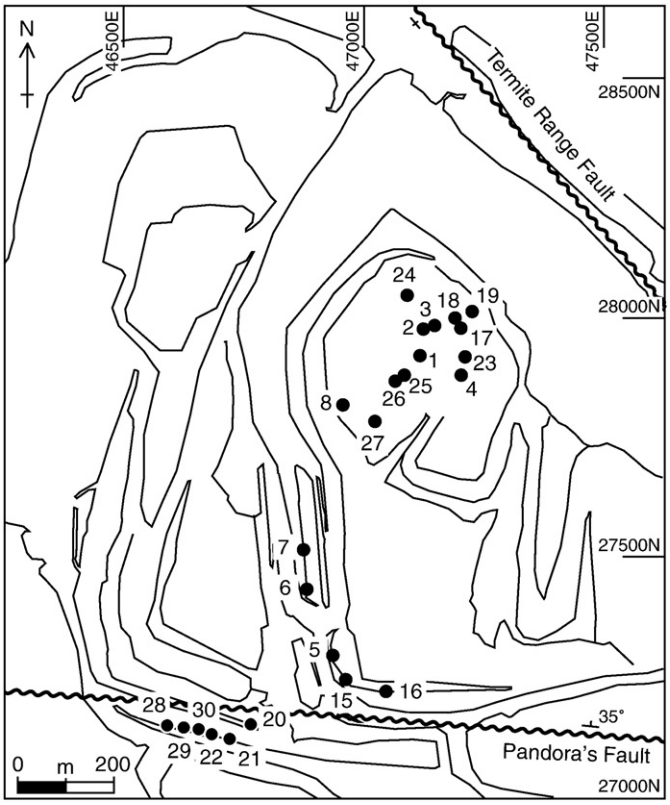



**Fig. 4.** Sampling site locations in the Century ore deposit with respect to the mine grid. The Termite Range fault is the approximate subsurface projection.

160 °C range calculated from vitrinite reflectance and illite crystallinity data for the deposit. Polito et al (2006) concluded that mineralization likely formed at a depth of 1900 to 3100 m. According to galena Pb/Pb model ages, the main ore-stage of mineralization at Century formed at ~1575 Ma (Carr et al., 1996). Ord et al. (2002) quoted 1570 ± 5 Ma as the refined age based on work by Carr (pers. comm. 2000). This age is ~25 Ma after the 1595 ± 6 Ma U–Pb zircon deposition age of the host shale sequence (Page et al., 2000). Broadbent et al. (1998) proposed that the Century deposit was formed 800 to 3000 m below the basin floor during basin inversion through late syndiagenetic replacement of organic-rich siltstone layers by fluids emanating from the nearby Termite Range fault (Fig. 3). However, other genetic models such as the SEDEX model have been proposed because many of the features of Century are similar to other SEDEX deposits for which both synsedimentary and syndiagenetic processes have been proposed (Large et al., 2005).

Regional lode deposits, such as Watson's Lode and the Silver King Lode (Fig. 3), occupy predominantly northeast-striking fault structures in a 10 × 20 km area to the west and southwest of the Termite Range fault (Broadbent et al., 1998). The fault structures cut folds that have been correlated with the $D_2$ event, and the fluid system during the $D_2$ event seems to have been responsible for the regional base-metal lode mineralization (Broadbent et al., 1998; Ord et al., 2002). The Watson's Lode is a vein- and breccia-style Zn–Pb–Ag deposit that occurs within the Lawn Supersequence of the Isa superbasin (Fig. 2) (Polito et al., 2006). Pb isotope model ages from galena at Watson's Lode range from 1575 to 1485 Ma (Carr et al., 2001, cited in Polito et al., 2006).

## 3. Methods and results

### 3.1. Sampling and initial measurements

Two to four oriented block samples were collected at 24 sites (Fig. 4), including 9 and 6 sites in mineralized zones of the north and south blocks, respectively, and 5 sites in footwall siltstone, 3 sites in hanging wall sandstones, and 1 site in hanging wall sideritic siltstone of the north block only (Table 1). The block samples were drilled perpendicular to bedding and then sliced into 6 to 16 core specimens, each 2.4 cm in diameter and ~2.20 cm long, per site in the paleomagnetic laboratory at the University of Windsor. The 333 specimens were stored for 1 month in a magnetically-shielded room with an ambient field of ~100 γ to allow their unstable and undesirable viscous remanent magnetization (VRM) components to

**Table 1**
Site mean characteristic remanent magnetization (ChRM) directions.

| Site | Lithology | Block | NE | NR | Strike (°) | Dip (°) | Dec. (°) | Inc. (°) | α95 (°) | *k* | Description (colour, grain size, lithology) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Unit 200 Mineralization | North | 4 | 7 | 200 | 8 | 276.9 | 84.5 | 5.7 | 71.3 | bk, vf, lam sh |
| 2 | Unit 430 Mineralization | North | 2 | 10 | 160 | 14 | 240.6 | 77.2 | 6.1 | 58.6 | bk, vf, strongly lam sh |
| 3 | Unit 450 Mineralization | North | 2 | 9 | 160 | 14 | 174.1 | 78.4 | 9.2 | 25.9 | dk gy, vf, lam sh |
| 4 | Unit 200 Mineralization | North | 1 | 11 | 110 | 12 | 271.6 | 74.9 | 8.4 | 31.2 | bk, vf, lam sh |
| 5 | Sandstone, hanging wall | North | 0 | 8 | 40 | 7 | 2.9 | −83.0 | 11.0 | 45.8 | lt gy, md, ss |
| 6 | Sandstone, hanging wall | North | 9 | 0 | 160 | 19 | 351.9 | −35.4 | 7.3 | 50.4 | lt gy, md, ss |
| 7 | Sandstone, hanging wall | North | 12 | 0 | 160 | 19 | 355.1 | −40.0 | 7.3 | 36.0 | lt gy, md, ss |
| 8 | Sideritic Siltstone, hanging wall | North | 13 | 0 | 204 | 9 | 150.8 | 76.9 | 5.8 | 52.3 | lt gy, f, lam stst |
| 15 | Siltstone, footwall | North | 3 | 3 | 78 | 48 | 260.1 | 62.0 | 17.1 | 18.4 | md gy, f, lam stst |
| 16 | Siltstone, footwall | North | 7 | 0 | 82 | 33 | 234.8 | 39.4 | 14.3 | 18.8 | md gy, f, lam stst |
| 17 | Siltstone, footwall | North | 8 | 3 | 148 | 19 | 278.2 | 35.9 | 11.2 | 17.9 | md gy, f, lam stst |
| 18 | Siltstone, footwall | North | 7 | 1 | 158 | 18 | 267.9 | 38.2 | 5.9 | 89.8 | md gy, f, lam stst |
| 19 | Siltstone, footwall | North | 0 | 5 | 152 | 16 | 273.4 | 60.9 | 4.4 | 653.9 | md gy, f, lam stst |
| 20 | Unit 410 Mineralization | South | 3 | 4 | 82 | 36 | 344.9 | 57.5 | 8.4 | 59.8 | bk, vf, strongly lam sh |
| 21 | Unit 160 Mineralization[a] | South | – | – | | | | | | | bk, vf, lam sh |
| 22 | Unit 170 Mineralization | South | 2 | 4 | 80 | 38 | 349.8 | 55.9 | 16.8 | 20.6 | md gy, vf, lam sh |
| 23 | Unit 430 Mineralization | North | 1 | 11 | 143 | 17 | 209.0 | 76.7 | 9.2 | 26.5 | bk, vf, strongly lam sh |
| 24 | Unit 410 Mineralization | North | 5 | 6 | 159 | 22 | 220.7 | 84.9 | 4.0 | 141.3 | bk, vf, strongly lam sh |
| 25 | Unit 160 Mineralization | North | 4 | 3 | 150 | 20 | 261.5 | 79.2 | 9.8 | 42.7 | dk gy, vf, lam sh |
| 26 | Unit 150 Mineralization | North | 5 | 7 | 161 | 14 | 272.0 | 79.3 | 5.9 | 57.7 | dk gy, vf, lam sh |
| 27 | Unit 200 Mineralization | North | 7 | 2 | 140 | 14 | 269.1 | 71.3 | 6.5 | 64.5 | bk, vf, lam sh |
| 28 | Unit 410 Mineralization | South | 4 | 1 | 89 | 32 | 333.1 | 58.9 | 2.5 | 972.4 | bk, vf, strongly lam sh |
| 29 | Unit 410 Mineralization | South | 2 | 8 | 76 | 27 | 325.0 | 68.6 | 4.7 | 120.0 | bk, vf, strongly lam sh |
| 30 | Unit 200 Mineralization[a] | South | 1 | 11 | 79 | 33 | 226.0 | 79.0 | 8.0 | 34.5 | dk gy, vf, strongly lam sh |

Notes: Mean declination (Dec.), inclination (Inc.) and radius of cone of 95% confidence (a95) in degrees and precision parameter (*k*) of Fisher (1953).
NE, specimen end points; NR, remagnetization circles; colour: bk, black; gy, gray; dk, dark; md, medium; lt, light; grain size: vf, very fine; f, fine; md, medium; lithology: sh, shale; stst, siltstone; ss, sandstone; lam, laminated.
[a] Sites omitted from average directions due to incoherent or aberrant directions.

substantially decay. All subsequent magnetic measurements were done without removing the specimens from the shielded room. The natural remanent magnetizations (NRM) of the specimens were measured on a 2G Enterprises 755R DC-SQUID magnetometer with a specimen sensitivity limit of $\sim 2\times10^{-6}$ A/m. The median NRM intensity of the mineralized specimens was $1.97\times10^{-4}$ A/m (quartiles: $Q_1 = 1.39\times10^{-4}$, $Q_3 = 2.81\times10^{-4}$) and that of the sandstone and siltstone specimens was $2.23\times10^{-4}$ A/m (quartiles: $Q_1 = 1.75\times10^{-4}$, $Q_3 = 3.35\times10^{-4}$).

### 3.2. Demagnetization

One typical specimen from each site was thermally demagnetized in 27 steps up to 680 °C using a Magnetic Measurements MMTD-80 oven with the thermal steps biased into the diagnostic unblocking temperature ranges of the common magnetic minerals *i.e.* goethite (80 to 140 °C), pyrrhotite (260 to 340 °C), magnetite (500 to 580 °C), and hematite (600 to 675 °C) (Dunlop and Özdemir, 1997). Most specimens, both from the mineralized zones and host rocks, show moderate to rapid decreases in remanence intensity up to ~140 °C, slower progressive decreases between 140 °C to ~380 °C, and then erratic increases above ~380 °C except for the sandstone specimens (Fig. 5). Two additional typical specimens from each site were alternating field (AF) demagnetized in 13 steps up to 120 mT using a Sapphire Instruments SI-4 demagnetizer. Based on the initial results, the remaining specimens were thermally demagnetized in 3 or 4 steps up to 250 °C and then AF demagnetized in 11 steps up to 120 mT (Fig. 6).

All remanence directions from mineralized zone and host rock specimens were plotted separately on Schmidt equal-area nets for each demagnetizing step using a computer program based on the Kamb (1959) method, which gives directional density contours in 5σ increments (e.g. Fig. 7). The NRM cluster was directed shallowly upward to the north, apparently recording mostly the present Earth's magnetic field (average 1900–2005 is $D = 5°$, $I = -48°$; IGRF-10, International Geomagnetic Reference Field). This cluster waned rapidly between 80 °C and 140 °C and started to migrate above 140 °C (Fig. 7), indicating that both VRM and modern goethite retained this initial direction. The mineralized specimens showed an ~70–80% intensity drop during thermal step demagnetization between NRM and 140 °C (Figs. 5, 6). Above 140 °C, the cluster migrated continually toward end points directed steeply downward to the northeast by ~70 mT (Fig. 7). The specimen ChRM directions were determined using both principal component (Kirschvink, 1980) and remagnetization circle analysis (Bailey and Halls, 1984), although remagnetization circle analysis proved more effective in determining the residual remanent components of residing magnetite and/or pyrrhotite in a majority of cases. The specimens from host rocks also showed a single cluster of NRM directions but started to migrate to differing directions above 140 °C.

Site and unit mean directions were calculated following Fisher (1953) (Table 1). The site mean ChRM directions for the mineralized specimens, excluding site 30, show distinct clustering of directions for both the north and south blocks (Fig. 8b). The mean direction for the mineralization in the north block is declination ($D$) = 247.2°, inclination ($I$) = 80.3°, number of sites ($N$) = 9, radius of cone of 95% confidence ($\alpha_{95}$) = 5.0° and precision parameter of Fisher (1953) ($k$) = 106. In the south block mineralized zone, site 21 failed to give coherent specimen ChRM directions, and site 30 gave a mean direction that was

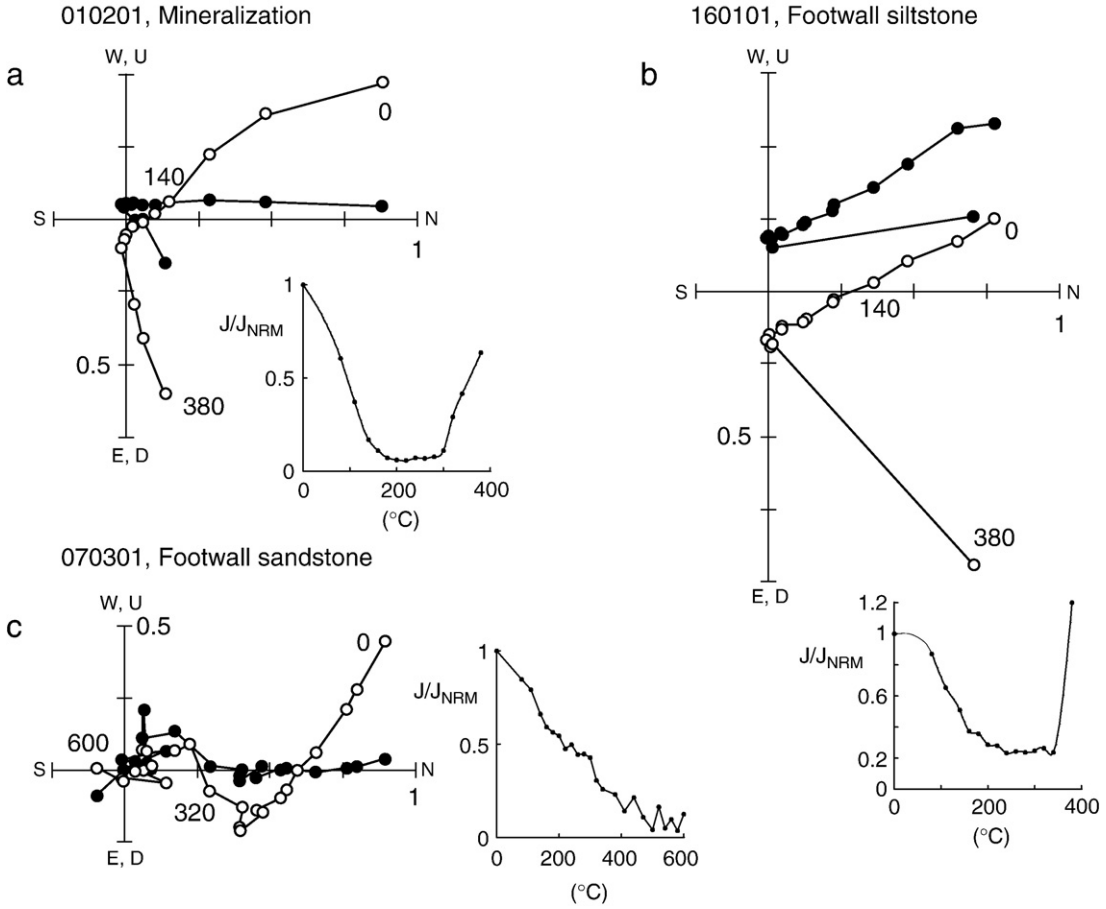



**Fig. 5.** Thermal step demagnetization of example specimens from sites 1, 7 and 16 with initial NRM intensities ($J_{NRM}$) of 3.45, 4.74 and $1.62\times10^{-4}$ A/m, respectively. Solid circles [open circles] denote projection on the horizontal [vertical] plane defined by the North (N), East (E), South (S) and West (W) [North (N), Down (D), South (S) and Up (U)] axes. Axial intensity values ($J$) are expressed as a ratio of the natural remanent magnetization (NRM) intensity ($J_{NRM}$). Some steps are labeled in °C.

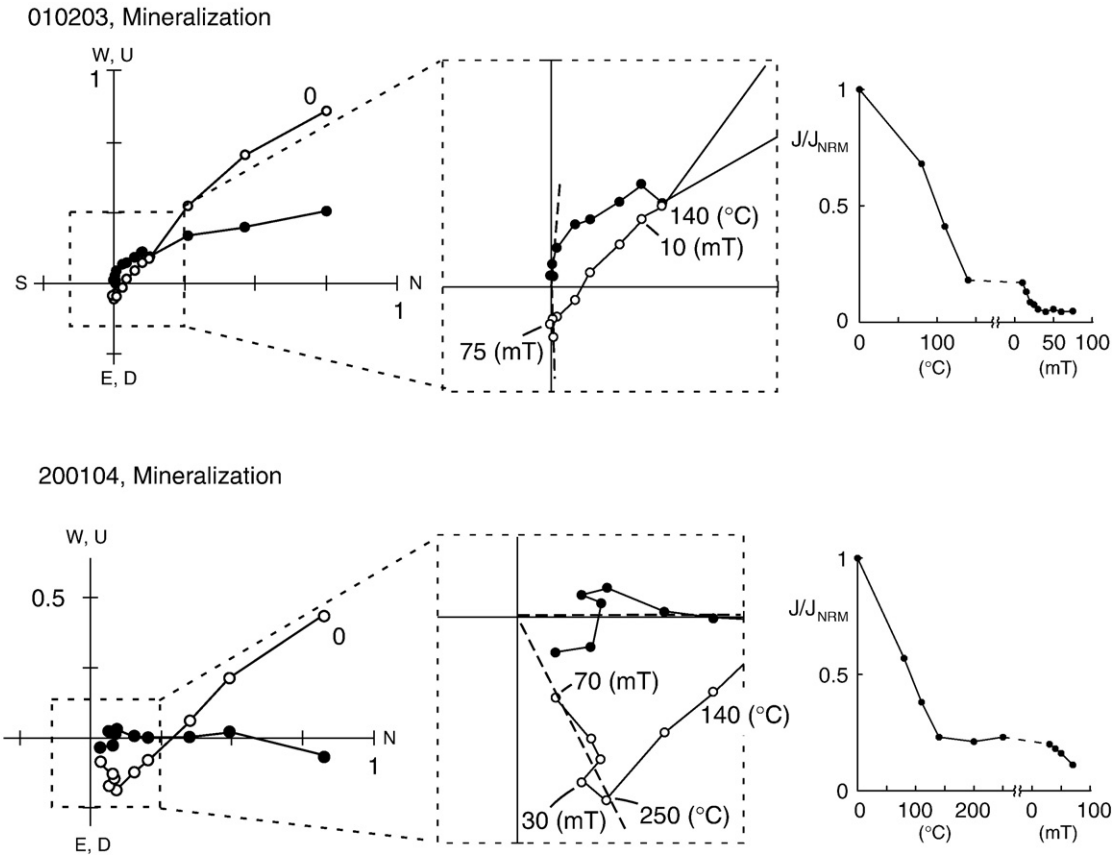



**Fig. 6.** Thermal (Th) and then alternating field (AF) step demagnetization of example specimens of mineralization from sites 1 to 20 with initial NRM intensities ($J_{NRM}$) of 3.05 and $1.57\times10^{-4}$ A/m, respectively. The plotting conventions are as in Fig. 5.

aberrant relative to the remaining four sites. Therefore the results of sites 21 and 30 were excluded from further analysis, giving a mean direction for the mineralization in the south block of $D=339.6°$, $I=60.5°$ ($N=4$, $\alpha_{95}=8.9°$, $k=108$) (Fig. 8b). Conversely, the site mean ChRM directions for the 4 hanging wall sandstones and siltstone sites ($D=351.4°$, $I=-59.1°$, $N=4$, $\alpha_{95}=29.6°$, $k=10.6$) and the 5 footwall siltstones sites ($D=262.6°$, $I=48.4°$, $N=5$, $\alpha_{95}=17.1°$, $k=21.0$), which are all in the north block only, are both poorly clustered and discordant with the ore block directions (Fig. 8a).

### 3.3. Fold test

The 13 accepted ChRM directions for the ore sites from the north and south blocks (Table 1) were compared using a paleomagnetic fold test to determine the timing of magnetization relative to tectonic deformation. Untilting the bedding causes the precision parameter $k$ to increase from 23.6 at 0% unfolding to 49.3 at 100% unfolding with the optimal bedding correction at $80\pm25\%$ ($2\sigma$) unfolding, and therefore the fold test is positive (Watson and Enkin, 1993; Enkin, 2003) (Fig. 9). For the maximum $k$ value at 80% unfolding, the mean ChRM direction for the 13 sites in mineralization is $D=249.1°$, $I=73.8°$ ($N=13$, $\alpha_{95}=5.7°$, $k=54.0$) (Fig. 8c) (Watson and Enkin 1993). This positive result shows that the ore ChRM either predates or mostly predates the $D_2$ deformation stage, indicating that the mineralization retains an essentially prefolding magnetization.

### 3.4. SIRM acquisition test

The magnetic minerals and their effective domain size were tested using saturation isothermal remanence (SIRM) analysis of 9 specimens *i.e.* 3 in south block and 3 in north block mineralization, and 1 each in hanging wall sandstone, hanging wall siltstone, and footwall siltstone of the north block. SIRM analyses were done using a Sapphire Instruments (SI-6) pulse magnetizer to magnetize each specimen in 13 direct current (DC) field steps up to 900 mT. The value of $0.9\times J_{900}$ is used as saturation. The specimens were subsequently AF demagnetized in 11 steps up to 140 mT (Fig. 10a,b). The rapid acquisition of SIRM to saturation at 200 mT for the hanging wall sandstone (Fig. 10b) is typical of multidomain (MD) magnetite or pyrrhotite. The rest of the specimens are not entirely saturated magnetically by ~300 mT and demagnetized moderately rapidly to low values, indicating that the main remanence carrier is single domain (SD) or pseudosingle-domain (PSD) magnetite or pyrrhotite. The continuously increasing SIRM acquisition above 300 mT to 900 mT indicates the presence of a minor high-coercivity mineral *i.e.* goethite and/or hematite.

Based on the SIRM acquisition and thermal step demagnetization tests, the dominant magnetic mineral in the sandstones is MD magnetite. Such MD magnetite carries a substantial VRM that easily changes direction to the ambient Earth's magnetic field to give an aberrant remanence direction. The AF and thermal demagnetization (Fig. 5) results show the presence of MD magnetite and goethite in the mineralized specimens. However, these components are thoroughly demagnetized by heating up to 250 °C so that the SD or PSD magnetite components, which carry the ChRM, can be isolated by thermal demagnetization to 250 °C and then AF step demagnetization.

Siderite is common in sedimentary carbonate rocks. Siderite may crystallize either at shallow or deep depths during the burial of the sediments and during diagenesis (e.g. Pye, 1984; Feltrin et al., 2007). At Century, zincian siderites formed either from recrystallization of diagenetic siderites or through addition to the mineralized sequence during hydrothermal alteration (Cooke et al., 2000). The spatial distribution of siderite at Century is controlled by the lithological character of the host sediments and is most abundant in siltstone

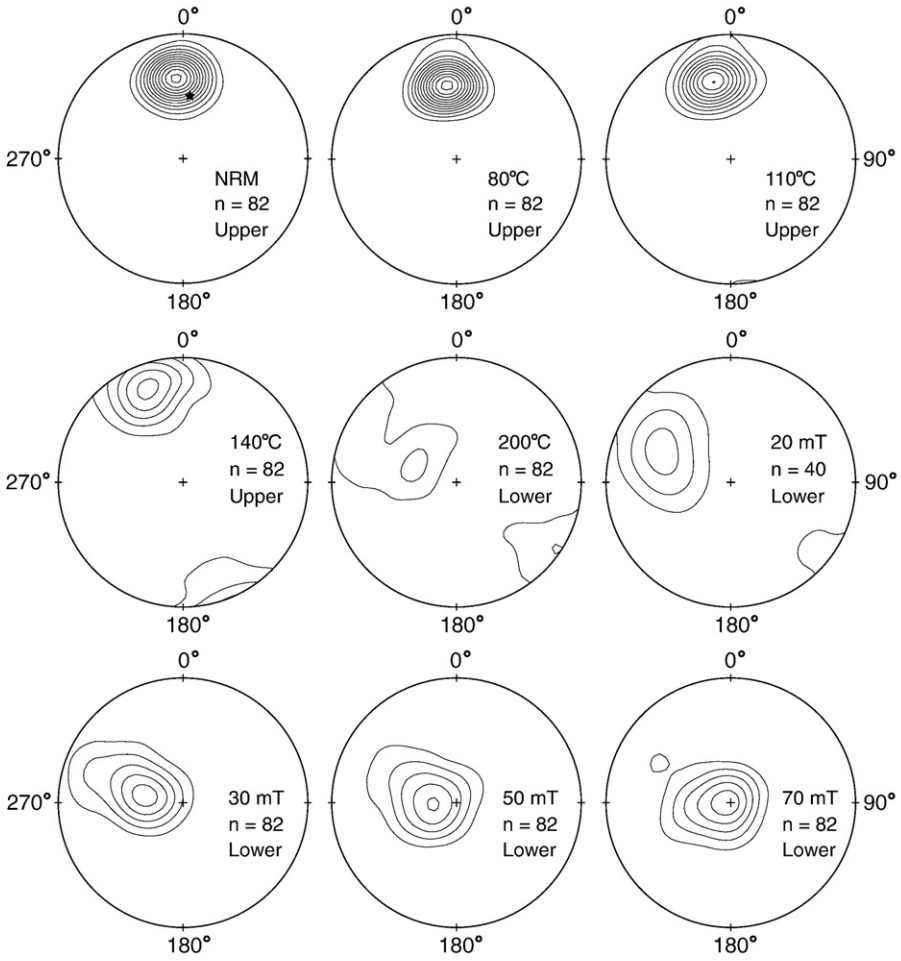



**Fig. 7.** Stereocontour plots for mineralized specimens from the north block on an equal-area projections of the upper (NRM and 80–140 °C) or lower (200 °C and 20–70 mT) hemisphere (Kamb, 1959) for thermal and then AF step demagnetization data. The number of specimens is *n*. The initial contour is at the 5σ level and subsequent increments increase by 5σ. The star shows the present Earth's magnetic field direction in the study area.

layers (Feltrin et al., 2007). Siderite is a paramagnetic iron mineral. However, natural siderites oxidize rapidly when exposed to air over periods of weeks to months, even at room temperature (Hus, 1990). In addition, siderite undergoes rapid oxidation to magnetite, maghemite or hematite above 300 °C (Fig. 5) (Dunlop and Özdemir, 1997). The magnetic minerals generated by oxidization during thermal step demagnetization acquire an unwanted chemical remanent magnetization that can overshadow the NRM of the specimens if sufficiently intense. The scattered ChRM directions in the siltstone specimens (Fig. 8a) are likely due to their abundant siderite. Although such sideritic specimens can be demagnetized by AF demagnetization, the Century specimens also contain goethite from the oxidation of the siderite that has coercivities of 10's of Teslas that cannot be AF demagnetized (Tauxe, 1998). Therefore, it is difficult to determine the

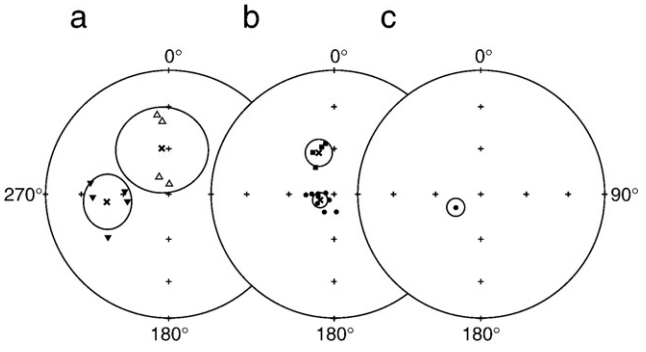



**Fig. 8.** An equal-area stereonet showing the site mean characteristic remanent magnetization (ChRM) directions with their cone of 95% confidence for: a) the hanging wall sandstones and siltstones (▼) and the footwall siltstones (▲); b) the mineralized zones from the north (●) and south (■) blocks; and, c) the unit mean ChRM direction for the mineralized zones with a 80% bedding correction.

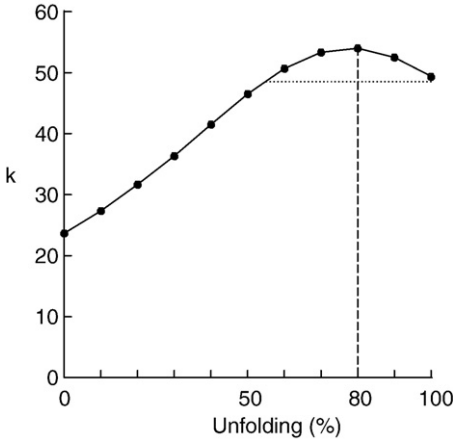



**Fig. 9.** Fold test (Watson and Enkin, 1993) for the 13 mineralized zone sites showing the change in the precision parameter (*k*) of Fisher (1953). Note that the best grouping of ChRM directions occurs at 80% unfolding (dashed line) with 95% confidence error (dotted line).

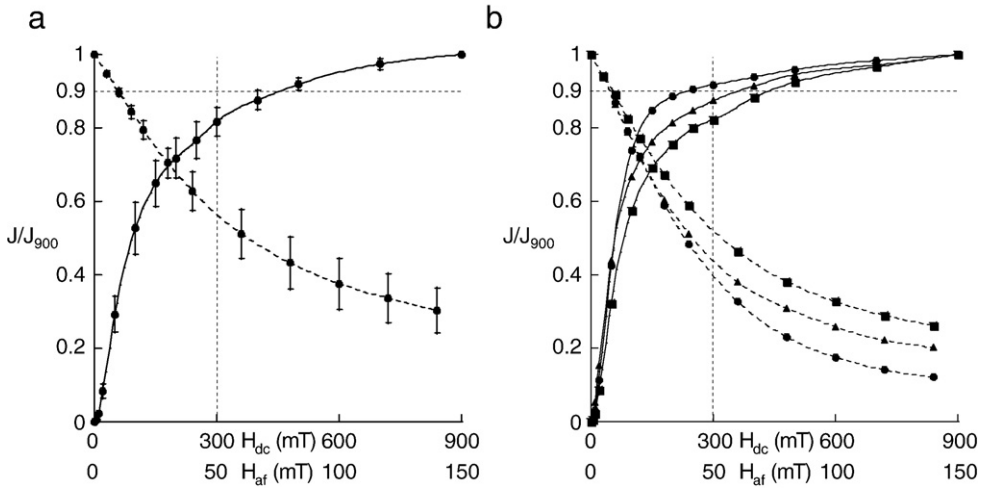



**Fig. 10.** Saturation isothermal remanent magnetization (SIRM) intensity for example specimens with their acquisition (solid lines; $H_{dc}$) and subsequent alternating field demagnetization curves (dashed lines; $H_{af}$). a) Average of 6 SIRM acquisition and decay curves with the standard deviation for 3 mineralized specimens from each of the north and south blocks; and b) SIRM acquisition and decay curves for one specimen each from the hanging wall sandstone (●), hanging wall siltstone (▲) and footwall siltstone (■).

primary remanence of the siltstone specimens at Century by either AF or thermal demagnetization.

### 3.5. Ore concentrate

It is important for paleomagnetic age dating of ore deposits to test that the ChRM is carried by the ore mineral(s) and/or ore-stage gangue minerals. For this purpose, 0.3 g of Zn and Pb concentrates and of tailings from the mine's mill were subjected to thermomagnetic analyses. Thermomagnetic curves were obtained in the temperature range from room temperature to 700 °C using a Petersen Instruments Magnetic Measurements Variable Field Translation Balance (MMVFTB) EM. A well-defined Curie temperature was observed around 320 °C for the Pb concentrate (Fig. 11a), which identifies pyrrhotite as the predominant magnetic carrier. Conversely, a continuous decrease of intensity up to ~400 °C is observed for the Zn concentrate that records the presence of titanomagnetite (Fig. 11c). The slower decreasing rate of intensity to 400 °C for the tailings than for the Zn concentrate and the remaining intensity after 600 °C indicate the presence of both titanomagnetite and hematite in tailings, although the hematite could have been generated by the oxidation of either magnetite or siderite during heating (Fig. 11c). The sudden increase in intensity around 450 °C in all three tests was caused by the transformation by oxidation of iron sulphide minerals such as pyrite and pyrrhotite to magnetite (Tarling, 1983; Dunlop and Özdemir, 1997). These results indicated that the ChRM for the mineralized zone is carried by inclusions of pyrrhotite in the galena mineralization and of titanomagnetite in both the sphalerite mineralization and the gangue.

## 4. Discussion

The mean ChRM direction of the ore sites after an optimal 80% bedding correction ($D=249.1°$, $I=73.8°$ ($N=13$, $\alpha_{95}=5.7°$, $k=54.0$) gives a pole position at 26.5°S, 107.0°E (dp = 9.2°, dm = 10.3°). This paleopole falls on the Mesoproterozoic portion of the apparent polar wander path (APWP) for northern Australia (Fig. 12), which is based tentatively by Idnurm (2000) on poorly age-constrained paleopoles from the Mt. Goldsworthy iron ores (MG3-I) and Yilgarn dykes (YA-I) in other cratons and on overprint paleopoles in the McArthur Basin (OP3) and Lawn Hill platform (OP3A). From fold tests, Idnurm (2000) interpreted both the OP3 and OP3A paleopoles to postdate the main $D_3$ deformation phase in the Mt. Isa Inlier. Connors and Page (1995) reported U–Pb zircon ages from pegmatites that set maximum and minimum age limits for $D_3$ of 1532 ± 7 Ma and 1480 ± 14 Ma, respectively. Mortimer et al. (1988) reported a Rb–Sr minimum age of 1549 ± 63 Ma for the Gawler Craton dykes (GB). The GB pole position (23°S, 86°E, $A_{95}=11°$; Idnurm and Giddings, 1988) is not significantly different from OP3 (25.9°S, 76.0°E, $A_{95}=8.1°$), indicating that the Australian cratons had likely been assembled by 1549 Ma so that the age of GB can be used for the northern Australian APWP. The paleomagnetic age of the Century mineralization, obtained by scaling along the APWP, is 1558 ± 4 Ma. Although the ore's paleopole includes OP3A, it precedes OP3A on the APWP and the positive fold test indicates that the mineralization predates the $D_2$ deformation stage. The 1558 ± 4 Ma age supports a late diagenetic replacement model for the genesis of the Century ore body. This 1558 ± 4 Ma age is slightly younger than the Pb–Pb model age of the 1570 ± 5 Ma (Carr, pers. comm., 2000, in Ord et al., 2002). The age difference could result from the poor age constraints on this part of the APWP. The regional Pb–Zn–Ag vein deposits display a wide range of isotope ratios with model ages between ~1585 Ma and ~1485 Ma (Richards, 1975; Large et al., 2005). Carr et al. (2001, cited in Large et al., 2005) interpreted these Pb–Pb data as mixed leads from at least three different events, *i.e.* a) a very early event (~1585 Ma) that could be synsedimentary; b) a major epigenetic event (~1575 Ma) that was responsible for the deposition of the bulk of the Century sulphide mineralization; and, c) a later epigenetic event (~1485 Ma) responsible for the Lawn Hill vein mineralization. Therefore, a younger age of 1570 ± 5 Ma for the Century mineralization could be the result of mixing with a later infiltrating Pb-bearing fluid (Feltrin et al., 2007). Since the positive paleomagnetic fold test results indicate that the mineralization retains a primary remanence and acquired the remanence before the $D_2$ deformation stage, it is highly unlikely that the Century mineralization was impacted by the later fluid. Therefore, the observed paleomagnetic age, which shows good agreement with the Pb–Pb age, suggests that the Pb–Pb galena ratios reflect also the mineralization age at Century.

The Mt. Isa–McArthur basin system hosts five giant SEDEX deposits (Century, HYC, Mt. Isa, George Fisher and Hilton) which exhibit similar morphological, textural and chemical features (Large et al., 2005). Symons (2007) reported paleomagnetic results for the HYC Zn–Pb deposit where diagenetic and metamorphic overprints are minimal, and found that the ChRM of both mineralized zone and host rocks gave a postfolding direction with an age that dictates an epigenetic origin at a depth of ~800 m. Also, Lewchuk et al. (2004) found that the Red Dog Zn–Pb–Ag deposit in Alaska was epigenetic in origin. These two paleomagnetic studies, plus this study, suggest that the syngenetic and syndiagnetic models, which have been thought to be favourable for the origin of these black shale SEDEX deposits (e.g. Leach et al., 2005), may not be correct.

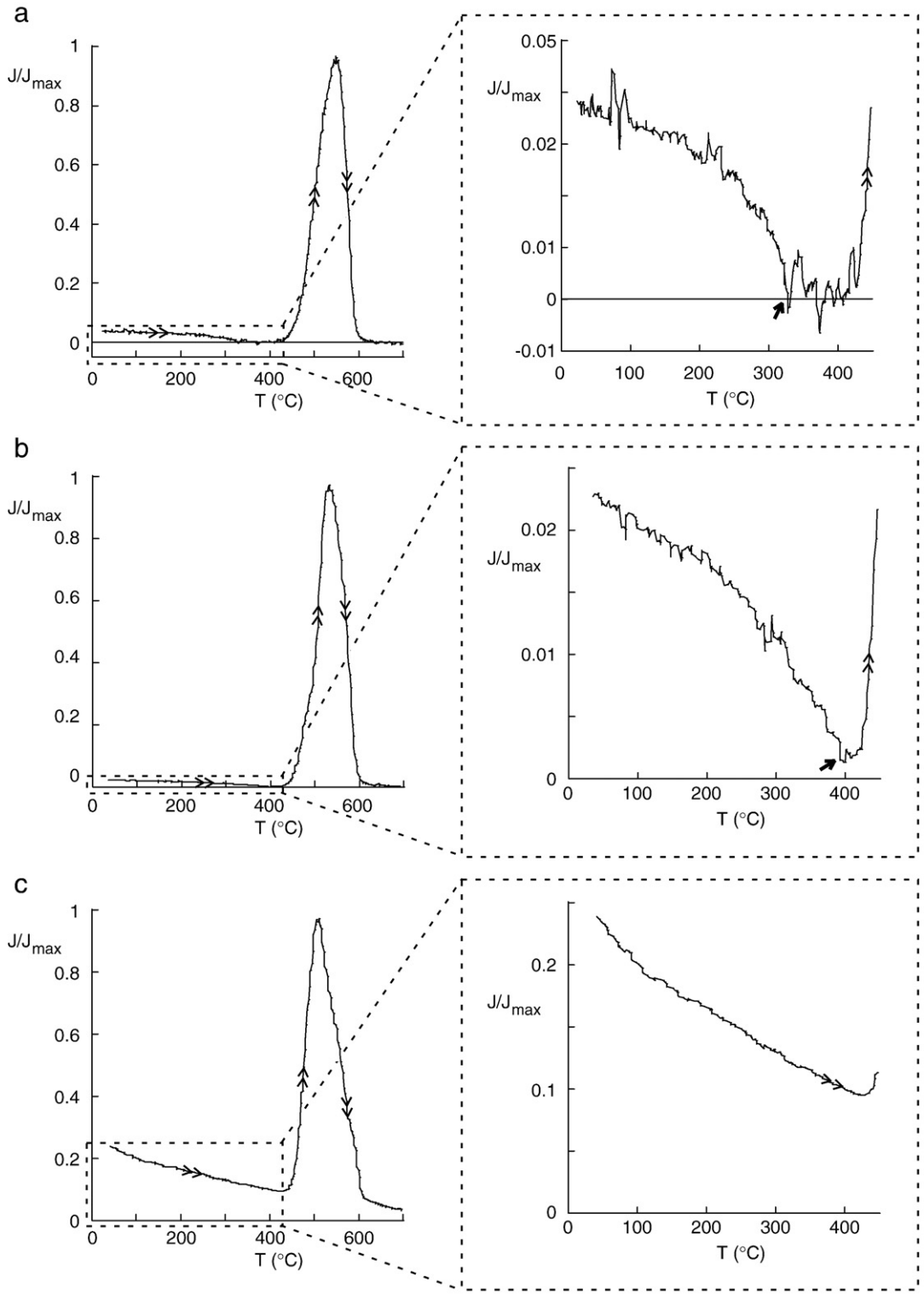



**Fig. 11.** Thermomagnetic heating curves with simple moving average ($n = 5$) for: a) Pb concentrate; b) Zn concentrate; and c) tailings. The curves were measured with the specimens in air. The expanded boxes show the heating curves up to 450 °C to spot the thermal changes before iron sulphide minerals are transformed to magnetite. The arrows show approximate Curie temperature. $J_{max}$ is for: a) $3.89\times10^{-2}$ $Am^2/kg$ in 175 mT direct field; b) $9.34\times10^{-2}$ $Am^2/kg$ in 182 mT direct field; and, c) $5.00\times10^{-2}$ $Am^2/kg$ in 169 mT direct field.

## 5. Conclusions

The positive fold test given by the ChRM directions from the mineralized zones shows that the ore ChRM predates the $D_2$ deformation stage of the Isan orogeny, indicating that the mineralization retains a primary remanence. Based on step demagnetization, rock magnetic tests and thermomagnetic analyses of ore, Pb and Zn concentrates and tailings, the main remanence carriers in the Century ore are SD or PSD titanomagnetite in sphalerite and gangue and pyrrhotite in galena along with modern goethite and/or hematite that mostly retain the present Earth's magnetic field direction. The pole position for the optimum 80% tilt-corrected unit mean ChRM direction for the Zn–Pb mineralization falls on the Mesoproterozoic portion of the APWP for northern Australia. The paleomagnetic age of

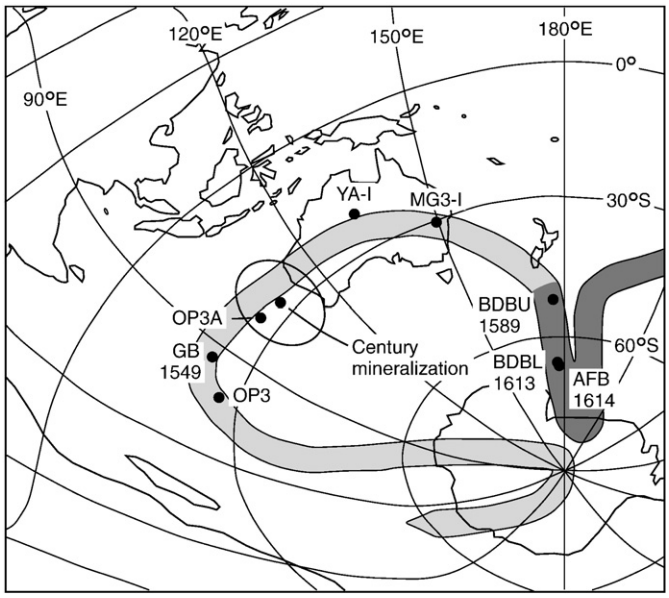



**Fig. 12.** Pole position for the 80% tilt-corrected unit mean ChRM direction of the mineralized zones with its oval of 95% confidence on the younger part of the apparent polar wander path (APWP) for northern Australia of Idnurm (2000). Parts of the APWP that are regarded as tentative are shown in light gray.

$1558 \pm 4$ Ma for the Century mineralization is close to the previously reported Pb–Pb galena model age of $1570 \pm 5$ Ma for the mineralization at Century, and both ages support a late diagenetic replacement model for the genesis of the Century ore body.

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